

SIMATS ENGINEERING

ASSESSMENT TOOL II — Visualization Analysis Report

Course: Data Handling and Visualization	Code: DSA06	CO: CO5
Duration: 180 min	Total Marks: 100	Reg No : 192524118 Name: R. AKshaya

COURSE OUTCOME COVERED

CO5: Analyze and evaluate visualization choices to communicate patterns, comparisons, relationships, and potential misleading effects through appropriate visualization methods. (BL4)

Activity Tool : Visualization Analysis Report

Weightage: 20%

Code of Conduct:

I _____R. Akshaya(192524118)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____R. AKshaya_____

Assessment Criteria

Criteria	Visualization Selection and Understanding 25%	Visualization Analysis and Interpretation 25%	Application of Visualization Principles 20%	Organization and Presentation 20%	Accuracy 10%	Clarity 5%	Total
Q1							
Q2							
Q3							
Q4							

Q5							
----	--	--	--	--	--	--	--

Visualization Analysis Report – Assessment Tool

R Programming Practical Questions with Datasets

Instructions: Answer all five questions. Use R programming to construct the required visualizations, analyze the patterns, compare visualization choices where requested, and identify potential visualization pitfalls. Use the given data exactly unless a transformation is explicitly requested.

Q1. Comparing Distributions – Histogram, Density and ECDF

A quality-control team records the following processing times (in minutes) for 20 randomly selected tasks. Using R, create a histogram and a density plot for the data. Also construct an empirical cumulative distribution function (ECDF). Compare the three visualizations and explain which aspects of the distribution—center, spread, skewness, and cumulative behavior—are communicated most clearly by each.

Data: Time = 18, 21, 22, 24, 25, 25, 27, 28, 29, 30, 31, 32, 33, 35, 36, 38, 41, 43, 46, 50 *Expected R tasks:*

histogram, density estimation, ECDF, appropriate labels/scales, comparison and interpretation

Q1: Processing Time Data

```
Time <- c(18,21,22,24,25,25,27,28,29,30,
```

```
31,32,33,35,36,38,41,43,46,50)
```

Histogram

```
hist(Time,
```

```
  main = "Histogram of Processing Times",
```

```
  xlab = "Processing Time (minutes)",
```

```
  ylab = "Frequency",
```

```
  col = "lightblue",
```

```
  border = "black")
```

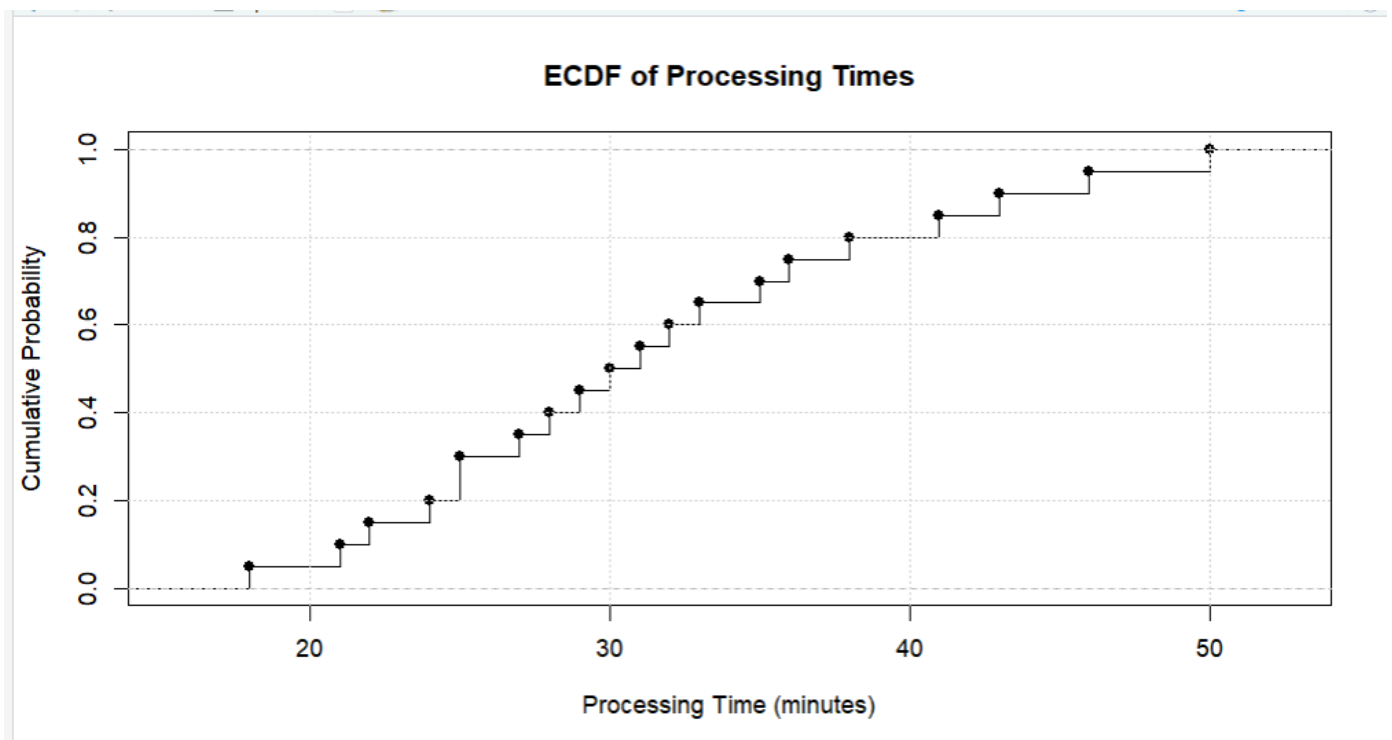
```
# Density Plot
```

```
plot(density(Time),  
  
     main = "Density Plot of Processing Times",  
  
     xlab = "Processing Time (minutes)",  
  
     ylab = "Density",  
  
     col = "blue",  
  
     lwd = 2)
```

```
# ECDF
```

```
plot(ecdf(Time),  
  
     main = "ECDF of Processing Times",  
  
     xlab = "Processing Time (minutes)",  
  
     ylab = "Cumulative Probability",  
  
     verticals = TRUE,  
  
     do.points = TRUE)
```

```
grid()
```



Interpretation

- The **histogram** shows the frequency of processing times in different intervals. Most observations are concentrated between approximately **25 and 38 minutes**.
- The **density plot** gives a smooth representation of the distribution. It shows the overall shape more clearly and indicates a **slight right skew** because some values extend toward 46–50 minutes.
- The **ECDF** shows the cumulative proportion of observations at or below each processing time.
- The processing times range from **18 to 50 minutes**, showing the overall spread.
- The center of the data is approximately around **30–32 minutes**.
- The histogram is useful for understanding **frequency, center and spread**.
- The density plot is useful for identifying the **shape and skewness**.
- The ECDF is most useful for understanding **cumulative behavior, probabilities and percentiles**

Q2. Q-Q Plot and Normality Analysis

A researcher measures the following response times (in seconds). Use R to create a normal Q-Q plot and a histogram/density visualization. Assess whether the observations are reasonably consistent with a normal distribution. Identify any departures from the reference line and explain what they indicate about the distribution.

Data: Response = 42, 45, 47, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 60, 61, 63, 66, 70, 76, 85

Expected R tasks: qqnorm()/qqline() or ggplot2 Q-Q plot, histogram/density plot, interpretation of deviations and outliers.

```
Response <- c(42,45,47,49,50,51,52,53,54,55,  
              56,57,58,60,61,63,66,70,76,85)
```

```
# Histogram
```

```
hist(Response,  
      main = "Histogram of Response Times",  
      xlab = "Response Time (seconds)",  
      ylab = "Frequency",  
      col = "lightgreen",  
      border = "black")
```

```
# Add density curve
```

```
lines(density(Response),  
      col = "red",  
      lwd = 2)
```

```
# Normal Q-Q Plot
```

```
qqnorm(Response,  
      main = "Normal Q-Q Plot of Response Times",  
      pch = 19)
```

```
qqline(Response,  
      col = "red",  
      lwd = 2)
```

Interpretation

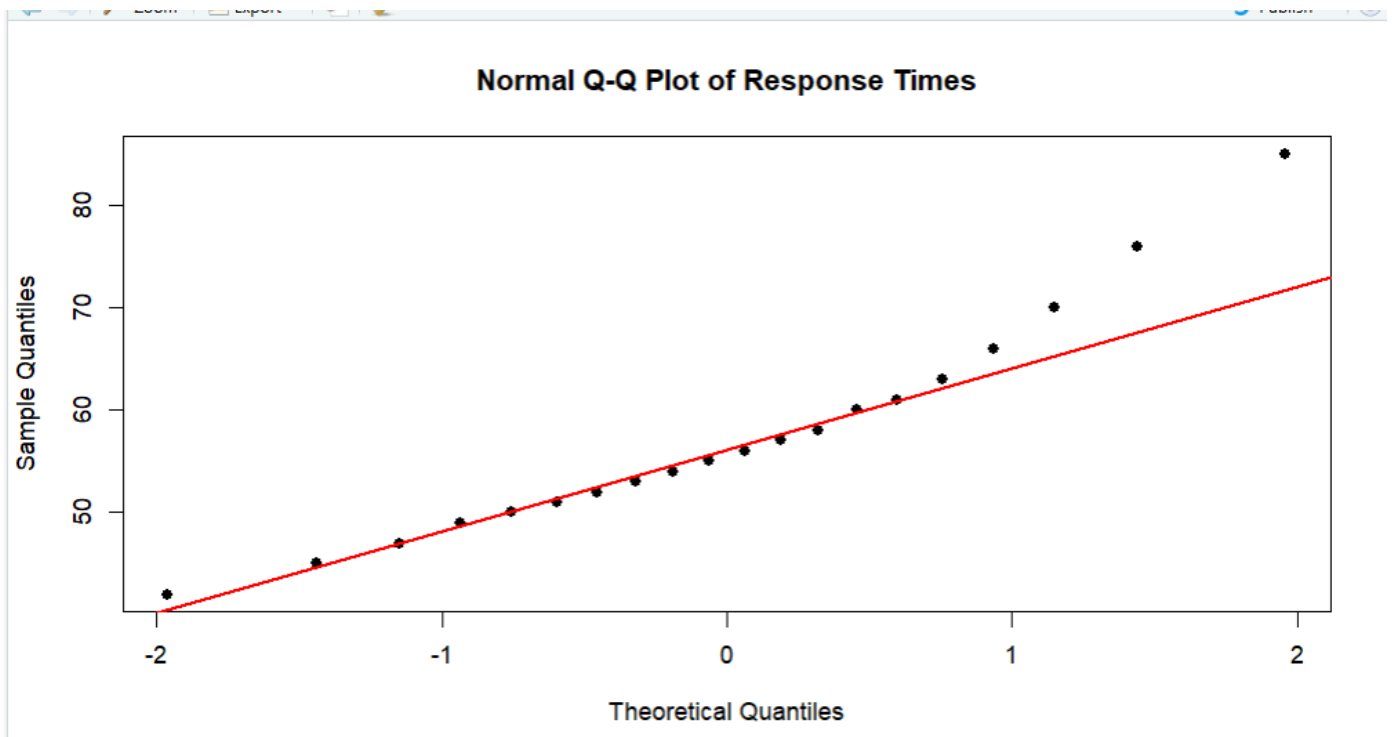
- The histogram shows that most response times are concentrated between approximately **45 and 66**

seconds.

- The values **70, 76 and 85** extend toward the higher end, creating a longer right tail.
- In a normal Q-Q plot, observations from a normal distribution should lie approximately along the reference line.
- Most points in the middle are reasonably close to the reference line.
- The points at the upper end move away from the line.
- This indicates **right-skewness** and a possible high-value outlier.
- Therefore, the data is **approximately normal in the central region but not perfectly normally distributed**.

Conclusion

The Q-Q plot shows that the observations are reasonably close to normal in the middle, but the upper-tail deviations indicate right-skewness and a possible high-value observation.



Q3. Visualizing Many Distributions at Once

Three groups of students have examination scores recorded below. Create a suitable visualization that compares the distributions of all three groups at once. Use boxplots and/or violin plots, and optionally overlay individual observations. Analyze differences in center, spread, and possible outliers. Explain why the selected visualization is more informative than displaying three separate plots.

Group A = 52,55,57,58,60,61,62,64,65,66,68,70,72,74,78;

Group B = 45,49,51,53,55,56,58,59,60,62,63,65,67,69,71;

Group C = 58,60,61,63,64,65,67,68,70,71,72,74,76,80,84

Expected R tasks: data reshaping, grouped boxplot/violin plot, optional jitter, comparison of distributions, interpretation.

```
Group_A <- c(52,55,57,58,60,61,62,64,65,66,68,70,72,74,78)
```

```
Group_B <- c(45,49,51,53,55,56,58,59,60,62,63,65,67,69,71)
```

```
Group_C <- c(58,60,61,63,64,65,67,68,70,71,72,74,76,80,84)
```

```
# Combine the data
```

```
Scores <- c(Group_A, Group_B, Group_C)
```

```
Group <- factor(c(rep("Group A", 15),  
                  rep("Group B", 15),  
                  rep("Group C", 15)))
```

```
# Boxplot
```

```
boxplot(Scores ~ Group,  
        main = "Comparison of Examination Scores",  
        xlab = "Student Group",  
        ylab = "Examination Score",  
        col = c("lightblue", "lightgreen", "lightpink"))
```

```
# Add individual observations
```

```
stripchart(Scores ~ Group,  
           vertical = TRUE,  
           method = "jitter",  
           pch = 19,  
           add = TRUE)
```

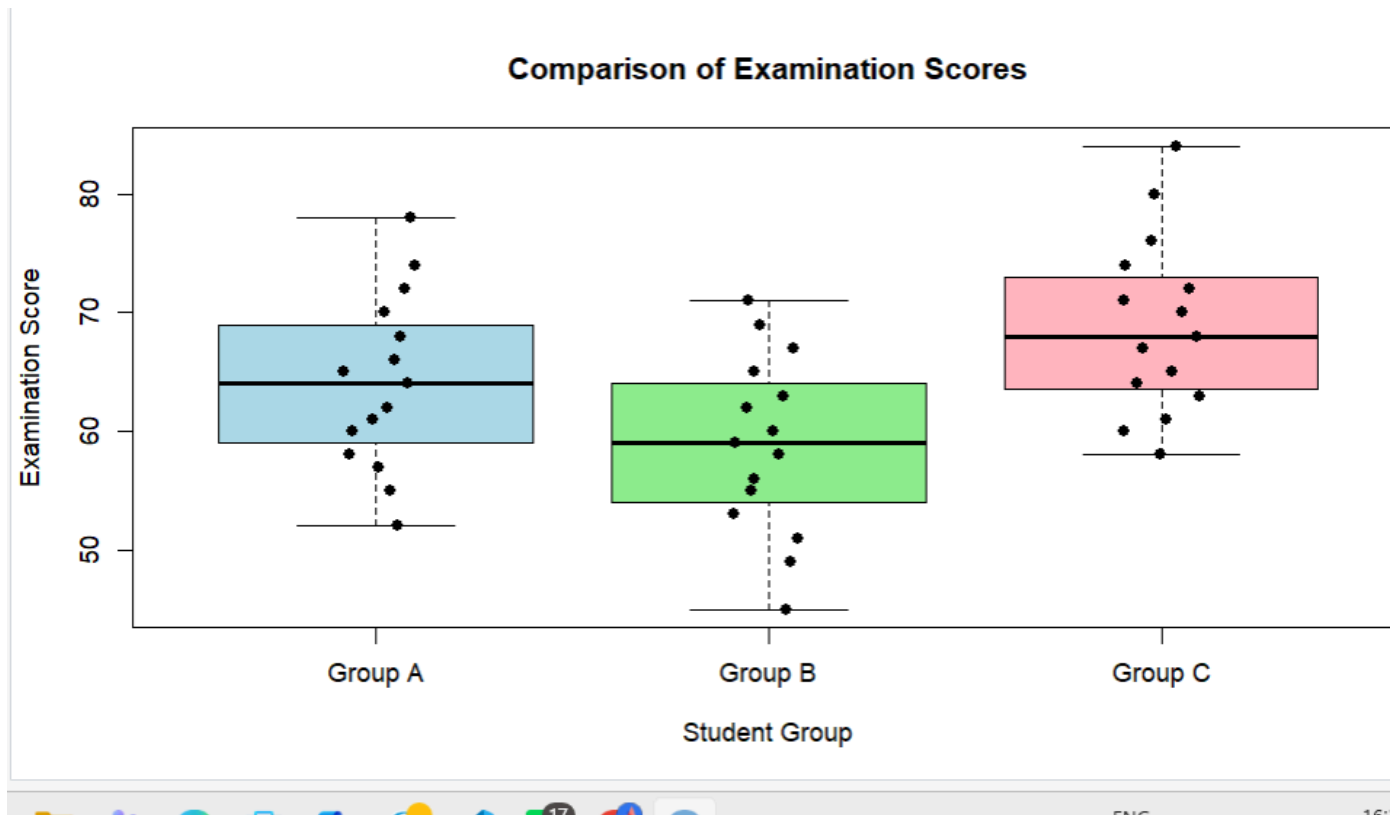
Interpretation

- **Group A** has a relatively high center, with scores ranging from **52 to 78**.
- **Group B** has the lowest center among the three groups. Its scores range from **45 to 71**.
- **Group C** has a high center and the widest upper range, extending up to **84**.
- Group B generally performs lower than Groups A and C.
- Groups A and C have generally higher scores.

- The individual points make it possible to see the actual observations in addition to the summary provided by the boxplot.
- There are no obvious extreme outliers based on the usual boxplot rule.

Why this visualization is useful

A grouped boxplot is more informative than three separate plots because all groups are displayed **side by side on the same scale**. This makes it easier to compare their centers, spreads, ranges and possible outliers immediately.



Q4. Relationships and Multivariate Visualization

A study records advertising expenditure, website visits, and sales for 15 observations. Create a scatterplot of advertising expenditure versus sales and use an appropriate visual encoding to incorporate website visits. Add a fitted trend line. Analyze the strength and direction of the relationship and explain whether the additional variable improves the interpretation or risks cluttering the visualization.

Advertising = 10,12,14,16,18,20,22,24,26,28,30,32,34,36,38;

Visits = 120,135,150,165,180,195,210,225,245,260,275,290,305,320,340;

Sales = 48,50,53,55,58,61,63,66,69,71,74,77,80,83,86

AT2 – Visualization Analysis Report | Data Handling and Visualization | DSA06

Expected R tasks: scatterplot, trend line, third-variable encoding, appropriate legend/scale, relationship analysis and critique.

```
Advertising <- c(10,12,14,16,18,20,22,24,26,28,
                30,32,34,36,38)
```

```
Visits <- c(120,135,150,165,180,195,210,225,245,260,
            275,290,305,320,340)
```

```
Sales <- c(48,50,53,55,58,61,63,66,69,71,  
          74,77,80,83,86)
```

```
# Scatterplot  
plot(Advertising, Sales,  
     main = "Advertising Expenditure vs Sales",  
     xlab = "Advertising Expenditure",  
     ylab = "Sales",  
     pch = 19,  
     cex = Visits / 100)
```

```
# Fit linear model  
model <- lm(Sales ~ Advertising)
```

```
# Add trend line  
abline(model,  
       col = "red",  
       lwd = 2)
```

```
# Calculate correlation  
cor(Advertising, Sales)
```

Interpretation

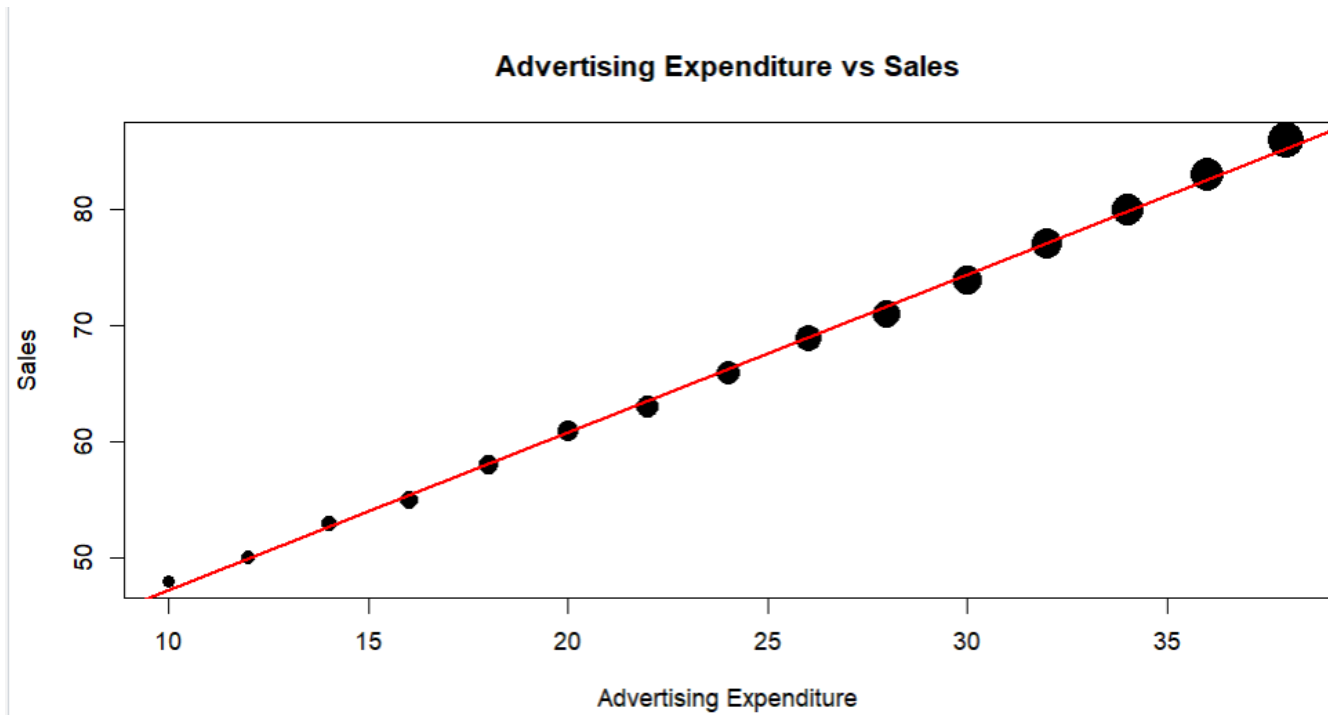
- The scatterplot shows a **strong positive relationship** between advertising expenditure and sales.
- As advertising expenditure increases, sales also increase.
- The points are closely arranged around an upward trend.
- The fitted trend line confirms this positive relationship.
- Website visits are represented using **point size**. Larger points indicate higher numbers of website visits.
- Therefore, the graph provides information about three variables without requiring three separate graphs.

Visualization critique

The additional variable, website visits, **improves the interpretation** because it gives more information about each observation.

However, too many visual encodings can make a graph difficult to read. For example, using different colors, shapes, sizes, labels and multiple trend lines simultaneously could create clutter.

Therefore, using **point size alone** for website visits is a reasonable choice.



Q5. Misleading Visualization and Redesign

A college reports the following pass percentages for seven departments. Create two visualizations: (1) a misleading version using a truncated y-axis and an inappropriate visual emphasis, and (2) a redesigned version using an appropriate baseline, ordered categories, and clear labels. Compare the two plots and explain how the first visualization can exaggerate differences and why the redesigned visualization communicates the data more honestly.

Department = CSE, ECE, EEE, MECH, CIVIL, AIDS, IT;
 Pass_Percentage = 91,93,92,89,90,95,94

Expected R tasks: bar chart with controlled y-axis, redesigned bar chart, ordering, labels, misleading-axis critique, accessibility and communication analysis.

```
Department <- c("CSE","ECE","EEE","MECH","CIVIL","AIDS","IT")
```

```
Pass_Percentage <- c(91,93,92,89,90,95,94)
```

```
barplot(Pass_Percentage,
  names.arg = Department,
  ylim = c(85,96),
  main = "Pass Percentage by Department - Misleading",
  xlab = "Department",
  ylab = "Pass Percentage (%)",
```

```
col = "orange",  
border = "black")
```

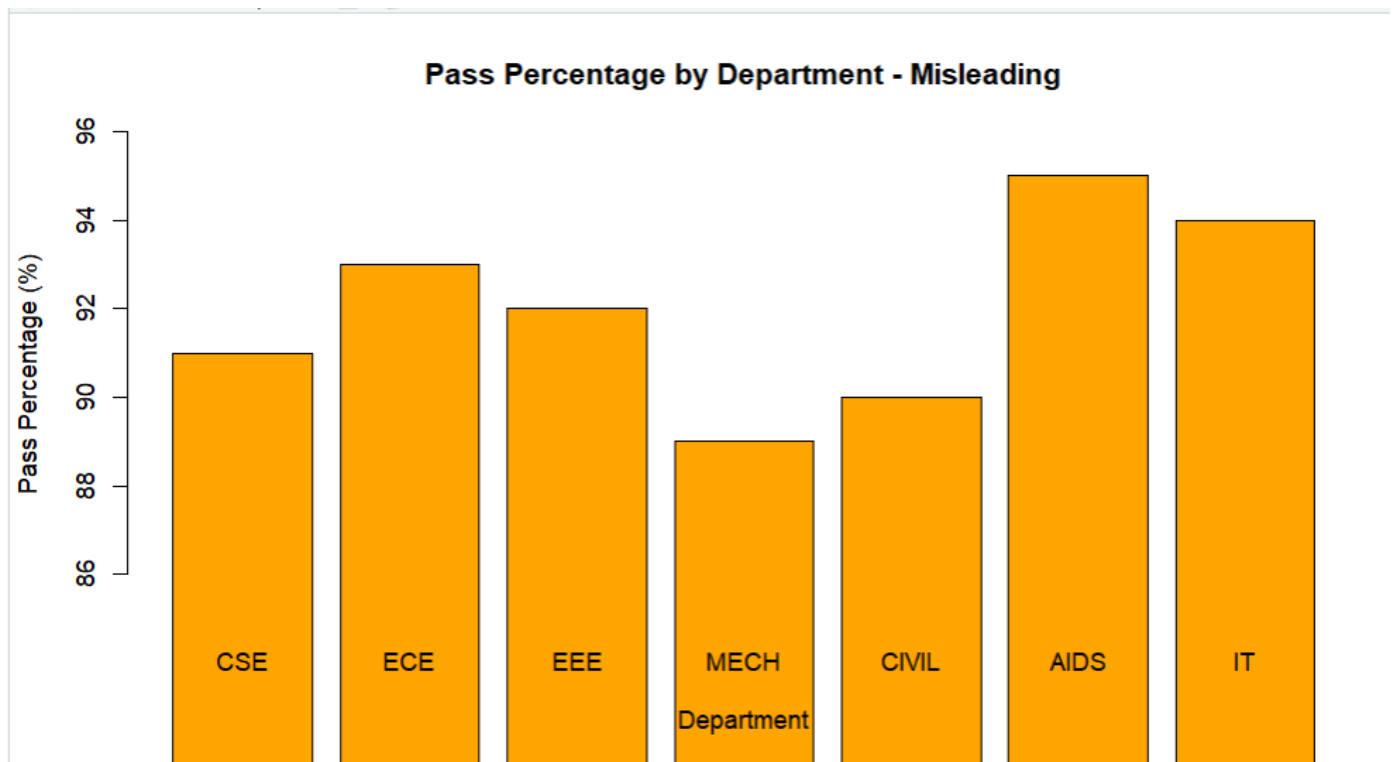
Interpretation

The graph starts the y-axis at **85% instead of 0%**.

The actual pass percentages range only from **89% to 95%**, so the difference between the lowest and highest department is only **6 percentage points**.

However, because the graph begins at 85%, the bars can make the differences appear much larger than they actually are.

This is a **misleading visualization** because



cause the visual size of the differences does not accurately represent their numerical size.

2. Redesigned Visualization

First, order the departments according to their pass percentage.

```
# Order the departments
```

```
order_index <- order(Pass_Percentage)
```

```
Department_ordered <- Department[order_index]
```

```
Pass_ordered <- Pass_Percentage[order_index]
```

```
# Redesigned bar chart
```

```
barplot(Pass_ordered,  
        names.arg = Department_ordered,  
        ylim = c(0,100),  
        main = "Pass Percentage by Department",  
        xlab = "Department",  
        ylab = "Pass Percentage (%)",  
        col = "skyblue",  
        border = "black")
```

```
# Add percentage labels
```

```
text(x = seq_along(Pass_ordered),  
     y = Pass_ordered + 3,  
     labels = paste0(Pass_ordered, "%"),  
     cex = 0.9)
```

Interpretation

The redesigned graph uses a **0% baseline**, which provides a more honest representation of the bar heights.

After ordering the categories:

- **MECH = 89%**
- **CIVIL = 90%**
- **CSE = 91%**
- **EEE = 92%**
- **ECE = 93%**
- **IT = 94%**
- **AIDS = 95%**

The graph clearly shows that the departments have **fairly similar pass percentages**.

Final comparison

The first visualization exaggerates the differences because its y-axis begins at 85%. The redesigned visualization uses a **0% baseline**, orders the departments, and adds clear percentage labels.

The redesigned graph therefore communicates the information **more honestly, clearly and accessibly**, while the first graph could cause viewers to believe that the differences between departments are much larger than they really are.

